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MRAMnet: a multi-region attention network with contrastive disentangled learning for diabetic retinopathy grading

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ABSTRACT

Diabetic retinopathy (DR) is a leading cause of preventable blindness, requiring accurate severity grading for timely clinical intervention. Current automated grading methods face two key challenges: inadequate capture of regional lesion correlations, and poor performance on underrepresented severe grades due to class imbalance. This study developed and validated MRAMnet, a novel deep learning framework addressing these limitations through multi-region attention mechanisms and contrastive disentangled learning. Using two public fundus image datasets — APTOS2019 (3,662 images) and EYEPACS (35,126 images) — MRAMnet partitions fundus images into quadrants, processed through a dual-branch architecture combining channel-spatial attention and transformer mechanisms. A Decoupled Supervised Contrastive Loss (DSCL) function balances feature learning across all severity grades. Evaluated via 5-fold cross-validation, MRAMnet achieved classification accuracy of 89.07% (APTOS2019) and 90.31% (EYEPACS), representing a 3–5% improvement over state-of-the-art methods. For severe grades, F1-score improvements of 8.7% (severe non-proliferative DR) and 6.5% (proliferative DR) were observed. MRAMnet demonstrates strong potential clinical utility in supporting DR screening programmes, particularly for sight-threatening cases requiring urgent referral.

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KEYWORDS
Diabetic retinopathy; retinal image analysis; convolutional neural networks; deep learning; automated grading

1. Introduction

Diabetic retinopathy (DR) is a complication of diabetes mellitus and is one of the leading causes of blindness in humans (Liu et al. 2019; Antonetti et al. 2021). There is a higher prevalence of DR in diabetic patients (Bose et al. 2023). In recent years, clinical studies have demonstrated the importance of accurate DR grading in classifying the severity of DR, which is used to improve treatments (such as photocoagulation (Stahl et al. 2022), vitrectomy (Antoszyk et al. 2020), and intraocular drug injections (Hashida and Nishida 2023)). DR grading can be categorised by the number, size, and type of retinal surface lesions in fundus images. The Clinical International Diabetic Retinopathy International Grading Criteria (Wilkinson et al. 2003), defines five stages of DR severity, including No DR, Mild Nonproliferative Diabetic Retinopathy (Mild NPDR), Moderate Nonproliferative Diabetic Retinopathy (Moderate NPDR), Severe Nonproliferative Diabetic Retinopathy (Severe NPDR) and Proliferative Diabetic Retinopathy (PDR). In the subclinical stage, the patient has no visible retinopathy. In the clinical stage, the earliest stage of DR is mild NPDR, which is characterised by microaneurysms (MA) occurring due to the leakage of tiny retinal vessels. The severity of DR then evolves further to moderate NPDR, in which MAs begin to increase, and other lesions (haemorrhages, exudates) develop. When severe NPDR develops, a large number of MAs, HMs, and EXs are spread across the retinal surface. A clear indicator of progression from NPDR to PDR is the growth of new blood vessels, which can result in severe vision loss. These physiological changes in retinopathy result in visual differences in fundus images, which can be effectively classified into the five stages of the DR severity scale.

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In recent years, deep learning methods have achieved remarkable success across diverse medical imaging tasks, including retinal disease detection (Banerjee et al. 2025; Singh et al. 2026), brain tumour diagnosis (Satushe et al. 2025), medical image segmentation for liver (Banerjee et al. 2025) and thyroid applications (Banerjee et al. 2025), and pulmonary disease classification (Singh et al. 2025). Building on these advances, considerable research has been conducted on the automatic detection of DR, with a particular focus on feature extraction and data enhancement (Lin 2018; He et al. 2020; Al-Antary and Arafa 2021). These DR classification methods have significantly improved performance, but they still have some shortcomings when compared to traditional datasets, such as ImageNet (Deng et al. 2009). Based on a systematic review of existing literature in automated DR grading, we identify three fundamental challenges that remain insufficiently addressed: (1) Regional lesion correlation modelling: DR lesions exhibit characteristic regional distribution patterns across different quadrants of the retina, and clinical protocols require ophthalmologists to systematically examine each retinal quadrant following the Early Treatment Diabetic Retinopathy Study (ETDRS) guidelines. However, existing deep learning methods predominantly employ global feature extraction that fails to capture inter-regional spatial relationships critical for distinguishing adjacent severity grades. (2) Class imbalance in long-tailed distributions: Clinical DR datasets are inherently imbalanced—in the APTOS2019 dataset, normal cases comprise approximately 49% of samples, while severe NPDR and PDR together account for less than 13%, leading to degraded detection of clinically critical severe cases (Ju et al. 2022). (3) Conflicting multi-objective optimisation: When simultaneously optimising for feature representation quality (via contrastive learning) and classification accuracy, conventional training strategies suffer from conflicting gradient directions, which is particularly detrimental for minority classes with limited gradient signals.

To address these identified challenges, this paper proposes MRAMnet, a novel Multi-Region Attention Module Network that leverages contrastive disentangled learning for robust DR grading. The main contributions of this study can be summarised as follows:

- We develop a Multi-Region Attention Module (MRAM) that explicitly models regional lesion correlations by partitioning fundus feature maps into four clinically meaningful quadrants. Unlike existing global attention mechanisms such as SE-Net and CBAM, MRAM processes each region through a novel dual-branch architecture combining channel-spatial attention for local refinement and transformer mechanisms for intra-region long-range dependencies. A cross-attention fusion mechanism enables bidirectional information exchange between branches, effectively capturing both fine-grained lesion characteristics and their regional distribution patterns critical for accurate DR grading.
- We introduce a contrastive disentangled learning framework implementing Decoupled Supervised Contrastive Loss (DSCL) that ensures class-balanced feature learning regardless of sample frequencies. By decoupling positive samples and assigning adaptive weights, DSCL guarantees that the gradient contribution ratio between majority and minority classes remains independent of class frequencies, directly mitigating the long-tailed distribution problem and enhancing generalisability for underrepresented severe DR grades.
- We implement a gradient projection mechanism within the contrastive disentangled learning framework that detects and resolves conflicting gradients between the DSCL-based contrastive branch and the ArcFace-based classification branch during backpropagation. When the cosine similarity between the two gradient vectors is negative, the contrastive gradient is projected onto the orthogonal complement of the classification gradient, ensuring that contrastive loss optimisation does not disturb learned classification boundaries.

The experimental results demonstrate that the proposed MRAMnet for DR classification achieved an accuracy of 89.07%, an F1 index of 0.891, and a top-2 accuracy of 98.6% in the APTOS dataset under 5-fold cross-validation. The accuracy on the Kaggle EYEPACS database reached 90.31%. These findings establish MRAMnet as a highly competitive system for five-stage DR grading, particularly excelling in the identification of underrepresented severe DR grades that present the greatest clinical challenges.

Comprehensive comparisons with recent state-of-the-art methods (2022–2026), including hybrid CNN–Transformer architectures, contrastive learning-based approaches, and multi-region processing methods, further validate the effectiveness of our approach.

The remainder of the paper is organised as follows. In Chapter 2, the state of research related to DR grading is discussed. In Chapter 3, specific details are described. In Chapter 4, the relevant datasets are presented and analysed. In Chapter 5, an evaluation of the proposed MRAMnet is presented to validate its performance. Finally, we draw conclusions regarding the proposed MRAMnet and discuss related future work in Chapter 6.

2. Related work

Currently, there are two principal approaches to the classification of DR: those based on traditional methods and those employing artificial intelligence. Traditional methods for DR classification require the design of manual features and classification of DR classes using generic classifiers or their variants, such as Support Vector Machines (SVM) and Random Forests (RF). Ramasamy et al. (2023) extracted features and classified different retinal fundus images from different retinal fundus images using an adaptive neural network-based Gaussian Laplace classification algorithm and compared them with the traditional BPN algorithm. Kashif et al. (Yaqoob et al. 2021) used a ResNet-based random forest classifier for retinal lesion classification. These traditional DR methods achieved good performance in the classification task but relied too much on a priori knowledge.

With the advancement of artificial intelligence techniques, Convolutional Neural Networks (CNN) have proven to be effective for many medical imaging tasks, including feature extraction, medical segmentation, and lesion detection. Goutam et al. (2022) optimised the problem of DR data imbalance and improved the feature extraction of tiny lesions by proposing a new categorical attention block (CAB) and global attention block (GAB). It can be seen that researchers have done a lot of work on feature processing and selection in CNN-based networks. Shoukat et al. (2023) processed fundus images in the grey channel using ResNet-50 as the architecture and used data augmentation to create large datasets to train convolutional network models to aid in the diagnosis of early glaucoma. Z. et al. (Ullah et al. 2023) proposed a semi-supervised learning method using U-net for the architecture of multi-decoder network trained on unlabelled datasets to improve segmentation performance. K. et al. (Parthiban and Kamarasan 2022) designed a chicken flock optimised (EN-CSOWNN) model by merging the wavelet transform method and EfficientNet The technique used U-net based segmentation method designed to determine the lesion region in fundus image for accurate classification. Deep convolutional network models led by ResNet (He et al. 2015), U-net (Ronneberger), and EfficientNet (Tan and Le 2019) have achieved excellent results in the field of medical image processing. Related research revolves around data enhancement methods and improving the network's ability for fine-grained representation. However, convolutional neural networks inherently prioritise local feature representations through their limited receptive fields, restricting their capacity for global contextual association. This limitation is particularly problematic for DR grading, where lesion distribution patterns across different retinal regions serve as key diagnostic indicators. Moreover, existing CNN-based DR methods process fundus images globally without explicitly modelling the regional correlations between lesions in different anatomical quadrants, a practice that is standard in clinical ophthalmological examination protocols.

With the introduction of the transformer, which has achieved great success in the field of natural language, the Vision Transformer applies the mechanism of multi-attention to the field of image processing and achieves the global representation of images by expanding them into one-dimensional vectors through embedding. With the emergence of ViT, owing to the secondary complexity problem of self-attention, which is the main component of the transformer, various ViT variant models have been developed one after another, such as the Swin Transformer (Liu et al. 2021), DeiT (Touvron et al. 2020), PVT (Wang et al. 2021), and DETR (Carion et al. 2020). DETR fuses the CNN and Transformer, which reduces the computational complexity limitations of pure ViT architectures, and is widely used in the medical image domain. However, ViT usually requires large medical images to learn a priori knowledge, and Yu et al. (2021a, 2021b) demonstrated that VIT is still effective on small fundus image datasets using an MIL head to assist VIT for diabetic retinopathy classification. Chen et al. (2021) introduced a hybrid encoder based on UNET, specifically a CNN and Transformer fusion, by applying the Transformer to the final layer of the encoder. By performing transformer feature extraction on the final layer of the encoder, the long-range modelling ability is enhanced, which is conducive to accurate localisation of segmentation. Cao et al. (2021)

constructed a jump-connected symmetric coder-decoder structure Swin-UNET using a Swin transformer block, explored the fusion method with a transformer as the backbone, and realised local-to-global self-attention. With the introduction of transformer, the fundus image model has a better representation, but its scarce data is still a challenge, Azizi et al. (2022) proposed the REMEDIS strategy, which uses a combination of large-scale supervised transfer learning and self-supervised learning without the need to adapt to specific tasks, to achieve data-efficient generalisation of medical AI. Zhou et al. (2023) developed a self-supervised-based model, RETFound, by learning from a large unlabelled retinal database and applying the learned features to a variety of downstream tasks for different diseases. Although the combination of CNN and Transformer compensates for the model's long-range relationship capture, it neglects the pathological relationships in the medical images themselves.

Recent advances in hybrid CNN-Transformer architectures have shown promising results for retinal disease diagnosis. Bhoopalan et al. (2025) proposed a Task-Optimised Vision Transformer (TOViT) specifically designed for DR detection in resource-constrained settings, integrating layer-wise learning rate scheduling and attention head tuning to enhance feature extraction while maintaining computational efficiency. Djoumessi et al. (2025) introduced an interpretable-by-design hybrid fully convolutional CNN-Transformer architecture that generates faithful and localised evidence maps directly reflecting the model's decision process. Singh et al. (2025) developed hybrid models combining deeper CNNs, Transformer encoders, and ensemble architectures for multi-disease retinal classification. More recently, CRA-Net (Zang and Ma 2024) employed cross-region attention mechanisms for fundus image analysis, utilising adaptive attention routing across spatial regions but without explicit clinical quadrant alignment. OcuViT (Ahmed and Uddin 2025) proposed an ocular-specific Vision Transformer pre-trained on large-scale ophthalmic data, demonstrating the benefit of domain-specific pre-training for retinal disease classification. LAT (Sun et al. 2021) developed a lesion-aware Transformer that incorporates lesion-specific tokens into the self-attention mechanism, enabling fine-grained feature learning for DR severity assessment.

While these studies highlight the effectiveness of CNN-Transformer fusion for retinal image analysis, they share a common limitation: the predominant focus on global feature integration without explicitly modelling the regional correlations between lesions across different retinal quadrants. Existing patch-based methods (e.g. ViT patch embeddings) divide images into uniform patches for computational purposes rather than clinically meaningful regions. Multi-region approaches such as CRA-Net learn attention routing adaptively but lack the clinical grounding of quadrant-based partitioning aligned with the ETDRS examination protocol. In contrast, MRAMnet is the first to integrate clinically motivated four-quadrant partitioning with a dual-branch cross-attention fusion architecture, explicitly mirroring the ophthalmological workflow for systematic DR assessment.

Addressing class imbalance remains a critical challenge in DR classification due to the inherent long-tailed distribution of clinical datasets. Recent works have explored diverse strategies for imbalanced medical image classification. Cai et al. (2024) proposed Balanced Parametric Contrastive learning (BPaCo) at MICCAI 2024, which employs across-batch class-averaging to balance gradient contributions from negative classes, achieving state-of-the-art performance on the APTOS2019 dataset. Chen et al. (2025) developed a Long-tailed Medical Diagnosis (LMD) framework featuring Relation-aware Representation Learning and Iterative Classifier Calibration using an Expectation-Maximisation approach. SSiT (Huang et al. 2024) introduced a self-supervised image Transformer pre-training strategy tailored for retinal image analysis, learning transferable representations that benefit downstream DR classification, particularly for underrepresented classes. DMCLNet (Guo et al. 2025x) proposed a dual-modality collaborative learning network that leverages cross-modal feature alignment with contrastive objectives to improve classification of rare DR grades. MediDRNet (Teng et al. 2024) designed a specialised architecture combining multi-scale feature extraction with class-balanced focal loss and feature re-weighting for DR grading, demonstrating improved sensitivity for severe DR grades. Xu et al. (2024) proposed a prior-guided attention fusion Transformer for DR lesion segmentation, integrating self-attention and cross-attention mechanisms with pre-trained vessel masks as prior knowledge. However, these approaches either focus solely on representation learning without considering regional lesion distribution, require additional annotations such as vessel segmentation masks, or treat class imbalance purely as a loss re-weighting problem without addressing the fundamental gradient conflict between representation learning and classification objectives. Our contrastive disentangled learning framework uniquely combines Decoupled Supervised Contrastive

Loss (DSCL) for class-balanced gradient optimisation with a gradient projection mechanism that resolves conflicting optimisation directions, providing a principled approach to simultaneously learning discriminative representations and accurate classifiers for long-tailed DR distributions.

Our approach addresses the above limitations through three synergistic innovations that distinguish MRAMnet from all existing methods. First, unlike global attention methods (SE-Net, CBAM), patch-based approaches (ViT, TMIL), or adaptively routed attention (CRA-Net), MRAMnet explicitly partitions feature maps into four clinically motivated quadrants aligned with the ETDRS examination protocol, processing each region through a dual-branch architecture that combines the inductive biases of CNN-based channel-spatial attention with the long-range modelling capability of Transformers. Second, unlike existing contrastive learning methods for DR (BPaCo, SSiT) that focus solely on representation learning, or loss re-weighting approaches (MediDRNet, Focal Loss) that address class imbalance at the loss level only, our contrastive disentangled learning framework integrates DSCL with gradient projection to simultaneously resolve class-frequency-dependent gradient inconsistency and multi-objective optimisation conflicts. Third, the cross-attention fusion mechanism in MRAM enables bidirectional information exchange between CNN and Transformer branches, a capability absent from existing hybrid architectures (TOViT, TransUNet, Swin-UNet) that rely on sequential or unidirectional feature integration. This comprehensive approach yields both improved overall classification accuracy and substantially enhanced detection of clinically critical underrepresented DR grades.

3. Method

This section describes the detailed architecture of MRAMnet. The overall framework is illustrated in Figure 1.

3.1. Network architecture overview

MRAMnet employs a dual-branch architecture with shared feature extraction backbone and specialised heads for contrastive learning and classification tasks. The network consists of three main components: a feature extraction backbone with integrated Multi-Region Attention Modules (MRAM), a contrastive

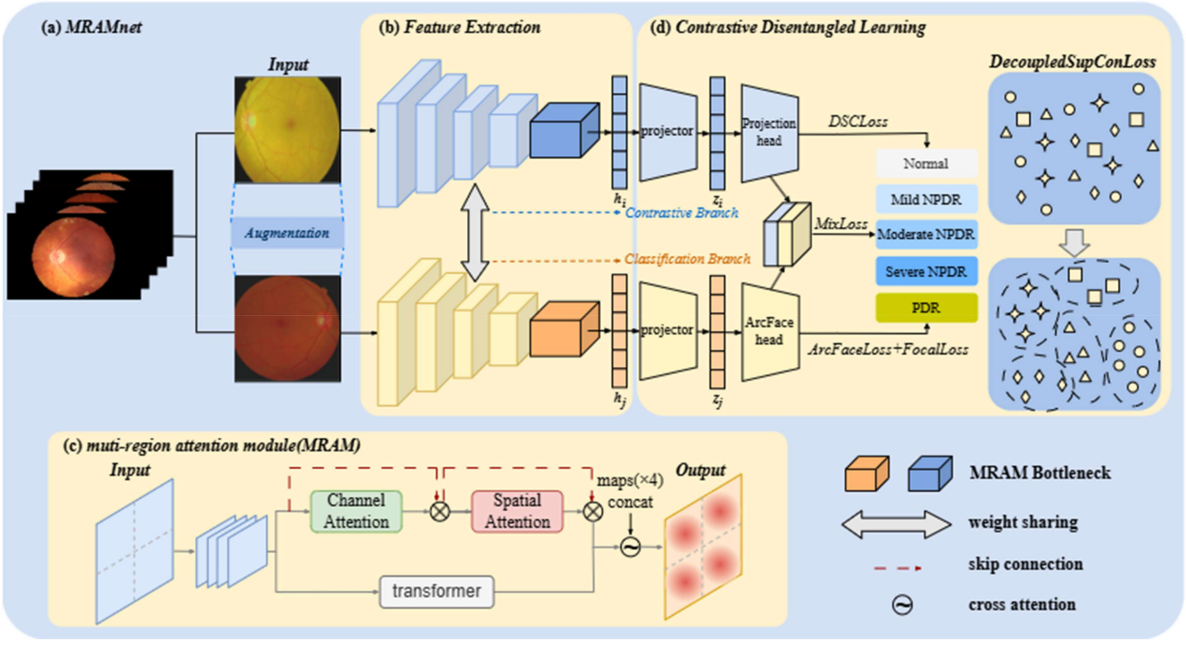


Figure 1. The overall architecture of MRAMnet for diabetic retinopathy grading. (a) Network overview with dual-branch design, (b) Feature extraction backbone with MRAM integration, (c) Multi-region attention module (MRAM) structure showing dual-branch processing with channel attention, spatial attention, and transformer with cross-attention fusion mechanisms, (d) Contrastive disentangled learning approach with decoupled supervised contrastive loss.

learning branch utilising Decoupled Supervised Contrastive Loss (DSCL), and a classification branch with ArcFace margin for enhanced class separation.

The feature extraction backbone is based on a modified ResNet-50 architecture. To enhance the network's ability to capture regional correlations and long-range dependencies critical for DR grading, we replace the standard 3×3 spatial convolutions in the final three bottleneck blocks (C5 layer) with our proposed Multi-Region Attention Module (MRAM). The original ResNet architecture follows a bottleneck design expressed as:

$$x_L = x_l + \sum_{i=L}^{L-1} F(x_i, W_i) \tag{1}$$

where W_i represents the weight of layer i, and each bottleneck consists of 1×1 convolution for dimension reduction, 3×3 convolution for feature extraction, and 1×1 convolution for dimension restoration. In our modified architecture, the MRAM-enhanced bottleneck preserves the skip connection while replacing the spatial convolution with the MRAM module. Table 1 details the architectural differences between standard ResNet-50 and our MRAMnet-50.

3.2. Feature extraction

3.2.1. Multi-region attention module (MRAM)

The MRAM represents the core innovation of our network, designed to enhance regional feature representation and capture correlations between lesions in different areas of the fundus image. The detailed architecture of MRAM is shown in Figure 2.

MRAM operates on the principle that diabetic retinopathy lesions exhibit strong regional characteristics, with their distribution and density across different quadrants of the retina being critical diagnostic indicators. Traditional convolutional operations struggle to capture these regional relationships efficiently, while pure transformer-based approaches lack the inductive biases necessary for processing medical images with limited training data.

While channel attention mechanisms (e.g. SE-Net) and spatial attention mechanisms (e.g. CBAM) have been extensively studied, our MRAM module differs fundamentally in three key aspects. First, Unlike existing attention mechanisms that operate on the entire feature map uniformly, MRAM explicitly partitions the feature space into four quadrants that correspond to the clinical practice of examining different retinal regions. This design choice is motivated by the observation that lesion distribution patterns across quadrants serve as important diagnostic indicators—severe DR typically exhibits widespread lesions across multiple regions, while mild cases show localised abnormalities. Second, Rather than the sequential application of channel and spatial attention as in CBAM, we employ parallel processing through two complementary branches. The channel-spatial attention branch provides strong inductive biases for local feature refinement, while the transformer branch captures long-range dependencies within each region.

Table 1. Comparison of ResNet-50 and MRAM-50 architectures in terms of layer configuration and parameter count.

Stage	output	ResNet-50	MRAMnet-50
c1	512×512	7×7, 64, stride 2	7×7, 64, stride 2
c2	256×256	3×3 max pool, stride 2	3×3 max pool, stride 2
		$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
c3	128×128	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$
c4	64×64	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$
c5	32×32	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ \textbf{MRAM}, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
# params.		25.5×10^6	23×10^6

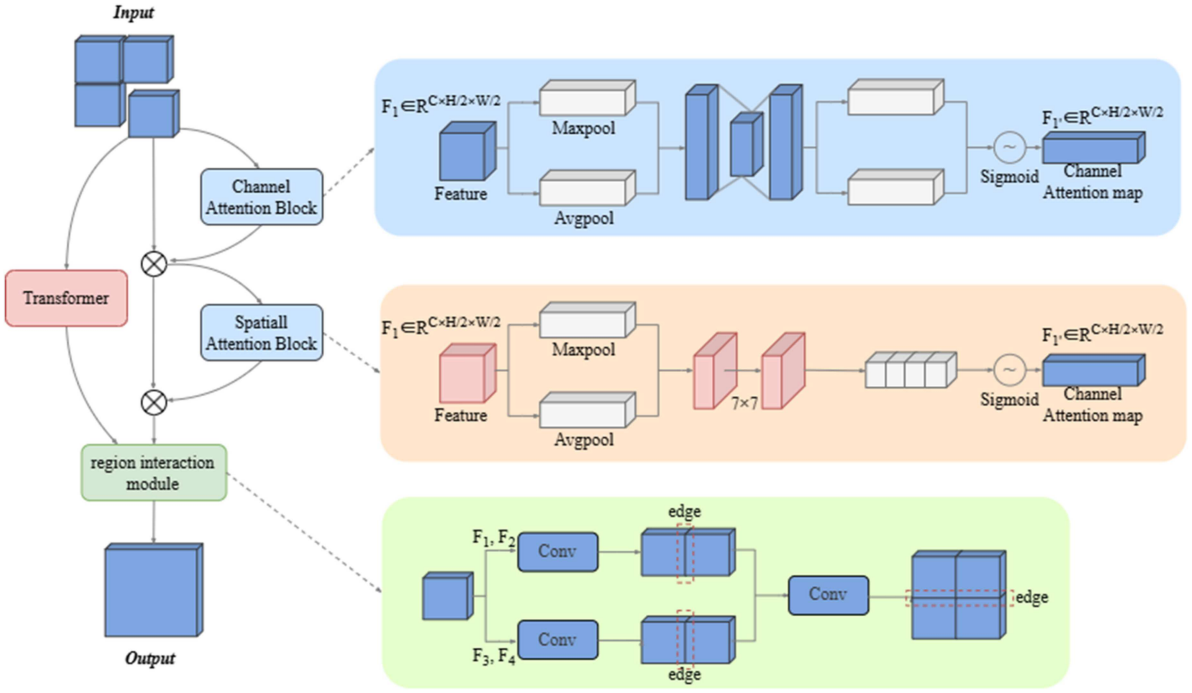


Figure 2. Basic architecture of the MRAM.

This parallel design enables simultaneous extraction of complementary feature representations without the information bottleneck inherent in sequential processing. Third, Our cross-attention fusion enables bidirectional information exchange between CNN-based and transformer-based feature representations. This addresses a key limitation of pure transformer approaches in medical imaging—the lack of appropriate inductive biases when training data is limited. By incorporating spatially-aware convolutional features into the transformer branch and vice versa, MRAM achieves robust performance even with the relatively small datasets typical in medical imaging applications.

Our MRAM architecture begins by partitioning the input feature map into four equal regions through reshape and permute operations. This division aligns with the clinical practice of examining fundus images by quadrants and enables specialised processing of each region. For a feature map of size $F \in \mathbb{R}^{B\times C\times H\times W}$, the partitioning creates four regions $F_i \in \mathbb{R}^{B\times C\times H/2\times W/2}$ where $i \in \{1,\ 2,\ 3,\ 4\}$.

Following partitioning, each region undergoes parallel processing through a dual-branch structure:

The first branch employs a sequential application of channel and spatial attention mechanisms with residual connections. The channel attention branch models inter-channel relationships by first generating a channel descriptor through global average pooling and global max pooling, followed by a shared MLP:

$$F_c^{avg} = GAP(F_i) \in \mathbb{R}^{B\times C\times 1\times 1} \tag{2}$$

$$F_c^{max} = GMP(F_i) \in \mathbb{R}^{B\times C\times 1\times 1} \tag{3}$$

$$M_c(F_i) = \sigma(MLP(F_c^{avg}) + MLP(F_c^{max})) \tag{4}$$

where σ denotes the sigmoid function. The channel-refined features are then processed by the spatial attention mechanism, which emphasises informative regions:

$$F_s^{avg} = AvgPool_{channel}(F_i \cdot M_c(F_i)) \tag{5}$$

$$F_s^{max} = MaxPool_{channel}(F_i \cdot M_c(F_i)) \tag{6}$$

$$M_s(F_i) = \sigma(Conv([F_s^{avg}, F_s^{max}])) \tag{7}$$

The output of this branch maintains the original spatial resolution while enhancing region-specific features:

$$F_i^{CA} = F_i \cdot M_c(F_i) + F_i(\text{residual connection}) \tag{8}$$

$$F_i^{CASA} = F_i^{CA} \cdot M_S(F_i^{CA}) + F_i^{CA}(\text{residual connection}) \tag{9}$$

The second branch processes each region through a transformer module that captures long-range dependencies. The feature map is flattened into a sequence and processed with multi-head self-attention:

$$F_i^{flat} = \text{Flatten}(F_i) \in \mathbb{R}^{B\times(H/2\cdot W/2)\times C} \tag{10}$$

$$F_i^{trans} = \text{Transformer}(F_i^{flat}) \tag{11}$$

The transformer utilises multi-head attention defined as:

$$\text{MHSA}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_h)W^o \tag{12}$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

$$= \text{softmax}((QW_i^Q)(KW_i^K)^T/\sqrt{d_k})VW_i^V \tag{13}$$

where Q, K, V are respectively Query, Key, ValueMatrix, W^Q, W^K, W^V is the corresponding weight matrix, W^O is the output weight matrix and d_k is the dimension of the key used to scale the dot product.

The critical aspect of our design is the cross-attention fusion between the two branches. This mechanism allows bidirectional information exchange, leveraging the complementary strengths of convolutional features (with strong inductive biases) and transformer features (with long-range dependency modelling). Specifically, the fusion is implemented through two cross-attention operations:

$$F_i^{C\to T} = \text{softmax}\left(\frac{F_i^{CASA}W_Q^{ct}\cdot(F_i^{trans}W_K^{ct})^T}{\sqrt{d_k}}\right)F_i^{trans}W_V^{ct} \tag{14}$$

$$F_i^{T\to C} = softmax\left(\frac{F_i^{trans}W_Q^{tc}\cdot(F_i^{CASA}W_K^{tc})^T}{\sqrt{d_k}}\right)F_i^{CASA}W_V^{tc} \tag{15}$$

where $F_i^{C\to T}$ denotes CNN-to-Transformer cross-attention (CNN features as queries attending to Transformer outputs) and $F_i^{T\to C}$ denotes Transformer-to-CNN cross-attention (Transformer features as queries attending to CNN outputs). W_Q, W_K, W_V are learnable projection matrices for queries, keys, and values respectively.

The final fused representation for each region is obtained through a learnable gating mechanism:

$$F_i^{fused} = \gamma \cdot F_i^{C\to T} + (1 - \gamma) \cdot F_i^{T\to C} \tag{16}$$

where γ is a learnable scalar parameter initialised to 0.5, allowing the network to adaptively balance the contributions from both cross-attention directions during training. This bidirectional cross-attention mechanism addresses the inherent limitations of transformers in medical image analysis—namely, their lack of appropriate inductive biases—by incorporating spatially-aware convolutional features. Simultaneously, it enhances the convolutional branch's ability to model non-local relationships critical for assessing lesion distributions.

After processing each region independently through the dual-branch architecture with cross-attention fusion, we recombine the region features to form a complete feature representation. Rather than simply concatenating the regions, we employ a Region Interaction Module (RIM) that enables information exchange between adjacent regions. The RIM is defined as:

$$\text{RIM}(F_a, F_b) = \text{ReLU}(\text{BN}(\text{Conv}_{1\times1}([F_a; F_b]))) \tag{17}$$

where $[F_a; F_b]$ denotes channel-wise concatenation of two region feature maps, $Conv_{1\times1}$ is a 1×1 convolution that reduces the channel dimension from $2C$ back to C, followed by batch normalisation (BN) and ReLU activation. This design allows the network to learn weighted combinations of adjacent region features rather than treating them equally.

The hierarchical aggregation proceeds as:

$$F_{top} = RIM\left(F_1^{fused}, F_2^{fused}\right) \tag{18}$$

$$F_{bottom} = RIM(F_3^{fused}, F_4^{fused}) \tag{19}$$

$$F_{complete} = RIM(F_{top}, F_{bottom}) \tag{20}$$

This hierarchical approach first combines horizontally adjacent quadrants (top-left with top-right, bottom-left with bottom-right), then vertically combines the resulting halves. This ordering preserves the spatial adjacency relationships between quadrants, reflecting the natural spatial structure of the retinal image.

Finally, the fused multi-region features undergo both average and max pooling, followed by a projection layer to generate the output feature map:

$$F_{pooled} = AvgPool(F_{complete}) + MaxPool(F_{complete}) \tag{21}$$

$$F_{output} = Projection(F_{pooled}) \tag{22}$$

This dual-pooling strategy captures both prevalent and distinctive features across regions, particularly important for identifying rare but critical lesions characteristic of progressive diabetic retinopathy.

3.2.2. MRAM in Bottleneck

The architectural innovation of MRAM module draws inspiration from the attention-based paradigm established in BoTNet (Srinivas et al. 2021), and replaces the spatial (3×3) convolution in the ResNet Bottleneck with a multi-region attention module to form the MRAM Bottleneck, as shown in Figure 3, this work extends this principle by implementing a multi-region attention operator that strategically substitutes the spatial convolution layer in conventional ResNet bottlenecks, thereby forming the proposed MRAM Bottleneck structure. This substitution enables simultaneous capture of cross-region dependencies while preserving the residual learning framework's stability.

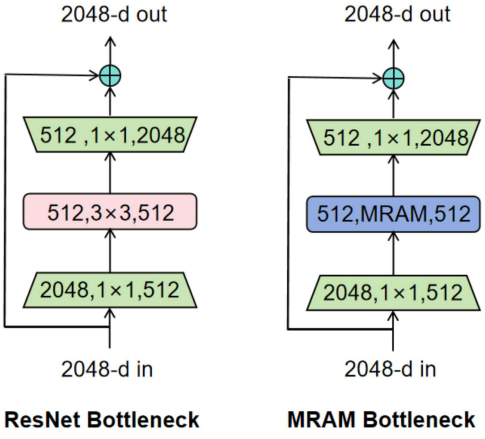


Figure 3. Left: ResNet Bottleneck structure. Right: MRAM Bottleneck structure.

3.3. Contrastive disentangled learning

To address the inherent class imbalance in DR datasets, where higher-grade DR images are typically underrepresented, we implement a contrastive disentangled learning approach. This branch projects the extracted features into a separate embedding space optimised for class discrimination through a projection head, which transforms the high-dimensional features extracted by the backbone (h_i) into a lower-dimensional space (z_i) more suitable for contrastive learning.

We implement the Decoupled Supervised Contrastive Loss (DSCL) approach (Xuan and Zhang 2024), which ensures more balanced optimisation of intra-class distances by decoupling positive samples and assigning different weights. DSCL can be expressed as:

$$\mathcal{L}_{\text{dscl}} = \frac{-1}{|\mathrm{P_i}| + 1} \sum_{\mathrm{Z_t} \in \{\mathrm{z_i^+} \cup \mathrm{P_i}\}} \log \frac{\exp \omega_{\mathrm{t}} (\mathrm{z_t} \cdot \mathrm{z_i} / \tau)}{\sum_{\mathrm{Z_m} \in \{\mathrm{z_i^+} \cup \mathrm{M}\}} \exp (\mathrm{z_m} \cdot \mathrm{z_i} / \tau)} \tag{23}$$

where z_i is the normalised feature vector for the anchor sample, z_i^+ is the positive sample from the same class as the anchor, P_i is the set of all other positives in the batch from the same class, M is the set of all negatives in the batch, τ is a temperature parameter (set to 0.07).

At the beginning of training, the L2 norm ratio of the two types of positive samples is:

$$\frac{\left| \frac{\delta L_{\text{dscl}}}{\delta z_i} \Big|_{z_i^+} \right|}{\sum_{z_{t \in P_i}} \left| \frac{\delta L_{\text{dscl}}}{\delta z_i} \Big|_{z_t} \right|_2} \approx \frac{\alpha}{1 - \alpha} \tag{24}$$

The term 'decoupled' in DSCL refers to the separation of the designated anchor-positive pair ($\mathrm{z_i}$, $\mathrm{z_i^+}$) from all other same-class positive samples $\mathrm{P_i}$ in the loss computation. In the standard Supervised Contrastive Loss (SupCon) (Khosla et al. 2020), all positive pairs are treated symmetrically, resulting in a gradient contribution ratio between majority and minority classes that scales linearly with class frequency $|\mathrm{P_k}|$. This creates a systematic bias where majority classes dominate the gradient updates, leading to poorly learned representations for minority classes.

DSCL resolves this by assigning an adaptive weight ω_t to each positive sample $\mathrm{z_t} \in \mathrm{P_i}$ and separately treating the anchor-positive pair. As shown in Equation (24), the L2 norm ratio of gradients from the anchor-positive pair versus other positives converges to $\alpha/(1 - \alpha)$ at optimality, which is controlled solely by the hyperparameter α and is independent of the number of samples per class. Setting $\alpha = 0.75$ yields a gradient ratio of 3:1 favoring the anchor-positive pair, ensuring that even classes with very few samples in a mini-batch receive proportionally balanced gradient updates.

The temperature parameter $\tau = 0.07$ controls the sharpness of the similarity distribution in the contrastive loss. Lower values of τ produce sharper distributions that emphasise hard negatives (samples close to the decision boundary), while higher values produce smoother distributions. We select $\tau = 0.07$ following the analysis in (Xuan and Zhang 2024), which demonstrates this value provides effective discrimination between similar and dissimilar pairs while maintaining training stability. The sensitivity of this choice is validated in our hyperparameter analysis (Section 5.6).

When DSCL converges, the optimality condition is approximately $P(Z_i^+ \mid Z_i) = \alpha$. This approach ensures that the gradient ratio is independent of the number of samples per class, resolving the inconsistency in feature learning between head and tail classes.

3.4. Classification branch with arcface

The classification branch transforms the extracted features (h_j) into classification logits with enhanced inter-class separability through ArcFace. The ArcFace loss can be expressed as:

$$\mathrm{L_{ArcFace}} = -\frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \log \frac{\mathrm{e}^{\mathrm{s} \cos (\theta_{\mathrm{y_i}} + \mathrm{m})}}{\mathrm{e}^{\mathrm{s} \cos (\theta_{\mathrm{y_i}} + \mathrm{m})} + \sum_{\mathrm{j} \neq \mathrm{y_i}} \mathrm{e}^{\mathrm{s} \cos \theta_{\mathrm{j}}}} \tag{25}$$

Where θ_{y_i} is the angle between the feature vector and the weight vector of the ground-truth class, mm m is the angular margin penalty (set to 0.5), and ss s is a scaling factor (set to 30). To address the class

imbalance problem, we implement a mixed loss combining ArcFace with Focal Loss, which helps the model focus on hard examples during training, particularly those from underrepresented classes.

3.5. Gradient projection for disentangled learning

MRAMnet's contrastive disentangled learning framework achieves hierarchical disentanglement of pathological features through a gradient projection mechanism that prevents conflicting optimisation objectives between the contrastive and classification branches. Unlike traditional multi-task learning that simply sums gradients, our approach detects and resolves gradient conflicts during backpropagation.

The gradient projection operates as follows. Let $g_{cls} = \nabla_\theta L_{cls}$ and $g_{con} = \nabla_\theta L_{con}$ denote the gradients of the classification and contrastive losses with respect to the shared backbone parameters θ. When the two gradient vectors point in conflicting directions (i.e. their cosine similarity is negative), the contrastive gradient is projected onto the orthogonal complement of the classification gradient:

$$g_{con}^{proj} = \begin{cases} g_{con} - \frac{g_{con} \cdot g_{cls}}{|g_{cls}|^2} g_{cls}, & \text{if } \cos(g_{con}, g_{cls}) < 0 \\ g_{con}, & \text{otherwise} \end{cases} \quad (26)$$

The total gradient update is then:

$$\frac{\partial L_{total}}{\partial \theta} = g_{cls} + \lambda_{con} \cdot g_{con}^{proj} \quad (27)$$

This projection is applied globally across all shared backbone parameters (i.e. the entire gradient vector of the shared feature extractor is treated as a single vector for the projection computation). Task-specific parameters (the contrastive projection head and the ArcFace classification head) receive only their respective gradients without projection.

The complete training procedure is detailed in Algorithm 1.

Algorithm 1. MRAMnet Training with Gradient Projection.

Input: Training data D, learning rate η, loss weight λ_con, max epochs E
Output: Trained model parameters θ, φ_cls, φ_con

Initialise backbone θ with ImageNet pre-trained ResNet-50 weights
Initialise classification head φ_cls and contrastive head φ_con randomly
for epoch = 1 to E **do**
  for each mini-batch (X, Y) in D **do**
    Generate augmented views: $X' = \text{Augment}(X)$
    Extract features: $h = f_\theta(X)$, $h' = f_\theta(X')$
    Compute contrastive embeddings: $z = g_\varphi con(h)$, $z' = g_\varphi con(h')$
    Compute classification logits: $\hat{y} = g_\varphi cls(h)$
    Compute $L_con = L_DSCL(z, z', Y)$
    Compute $L_cls = L_ArcFace(\hat{y}, Y) + L_Focal(\hat{y}, Y)$
    Compute $g_cls = \nabla_\theta\ L_cls$, $g_con = \nabla_\theta\ L_con$
    if $\cos(g_con, g_cls) < 0$ **then** ▷ *Conflicting gradients*
      $g_con\text{^proj} \leftarrow g_con - (g_con \cdot g_cls / \|g_cls\|^2) \cdot g_cls$ ▷ *Orthogonal projection*
    else
      $g_con\text{^proj} \leftarrow g_con$ ▷ *No conflict*
    end if
    Update $\theta \leftarrow \theta - \eta(g_cls + \lambda_con \cdot g_con\text{^proj})$ ▷ *Shared backbone*
    Update $\varphi_cls \leftarrow \varphi_cls - \eta \cdot \nabla_\varphi cls\ L_cls$ ▷ *Classification head*
    Update $\varphi_con \leftarrow \varphi_con - \eta \cdot \nabla_\varphi con\ L_con$ ▷ *Contrastive head*
  end for
  Adjust learning rate with cosine annealing schedule
  Evaluate on validation set; apply early stopping if no improvement
end for

3.6. Training strategy

The overall training procedure for MRAMnet involves data augmentation techniques to enhance the model's generalisation ability, including random horizontal flips (probability 0.5), vertical flips (probability 0.5), rotations (± 20°), random resized crops (scale 0.8–1.0), and colour jittering (brightness, contrast, and saturation adjustments of ± 0.1). The overall loss function combines contrastive and classification losses:

$$L_{total} = \lambda_{con} \cdot L_{contrastive} + (1 - \lambda_{con}) \cdot L_{classification} \quad (28)$$

where $\lambda_{con} = 0.5$ balances the contribution of contrastive and classification objectives. $L_{contrastive} = \mathcal{L}_{DSCL}$ (Equation (23)) and $L_{classification} = L_{ArcFace} + L_{Focal}$ (Equation 25). For optimisation, we use the AdamW optimiser (weight decay = 1×10^{-4}) with an initial learning rate of 1×10^{-3} and cosine annealing with warm restarts ($T_0 = 10$, $T_{mult} = 2$). The batch size is set to 16 and the model is trained for a maximum of 60 epochs with early stopping (patience = 10 epochs) based on validation performance to prevent overfitting. The backbone is initialised with ImageNet pre-trained ResNet-50 weights, while the contrastive projection head and classification head are initialised randomly. All input images are resized to 442 × 442 pixels and normalised using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

Through this comprehensive approach, MRAMnet effectively addresses the challenges of DR grading by combining regional attention mechanisms with contrastive disentangled learning, resulting in improved classification performance, particularly for underrepresented DR grades.

4. Dataset

Two publicly available fundus image datasets were used in this study: the EyePACS (Kaggle Diabetic Retinopathy Detection Competition 2015) and APTOS 2019 (APTOS 2019 blindness detection 2022) blindsight risk-challenge datasets. These two datasets contain fundus images from different patients, and the images are labelled with different stages of retinopathy, thus providing a rich sample for the retinopathy classification tasks in our study.

4.1. Original datasets

The EyePACS dataset, provided by the 2019 Kaggle Diabetic Retinopathy Detection Competition, includes over 35,000 high-resolution fundus images labelled with diabetic retinopathy severity, ranging from grades 0 to 4. The APTOS 2019 Blindness Detection dataset, provided by the Asia Pacific Tele-Ophthalmology Society (APTOS), contains over 3,600 fundus images similar to retinopathy severity, with a focus primarily on Asian patients.

4.2. Processed datasets

For each dataset, the images were divided into training (80%), validation (10%), and testing (10%) sets to ensure effective model training and generalisation. Each image filename contains patient and acquisition details, such as eye (left or right) and acquisition time.

To mitigate the risk of data leakage, patient-level separation was enforced during dataset splitting. For the EYEPACS dataset, we utilised the patient identifier embedded in each image filename to ensure that all images from the same patient (including both left and right eyes) were consistently assigned to the same fold during cross-validation and to the same partition during the initial train/validation/test split. This prevents the model from learning patient-specific characteristics rather than generalisable DR features. For the APTOS2019 dataset, explicit patient identifiers are not publicly available in the dataset metadata. We applied filename-based heuristic grouping to identify potential same-patient image pairs, but acknowledge that complete patient-level separation cannot be guaranteed for this dataset. This limitation is explicitly discussed in Section 5.10, and we note that our cross-dataset generalisation experiments (Section 5.7) provide an independent assessment of model performance free from any within-dataset data leakage concerns.

Ethical guidelines were followed to anonymize the datasets and protect patient privacy.

Before training, images undergo preprocessing (Figure 4):

(1) Mask contour cropping: Using the OTSU method to generate a binarized mask, the largest contour is detected and cropped to remove unnecessary background (Figure 4a).
(2) Image scaling: Cropped images are resized to 256 × 256 pixels using INTER_AREA interpolation.

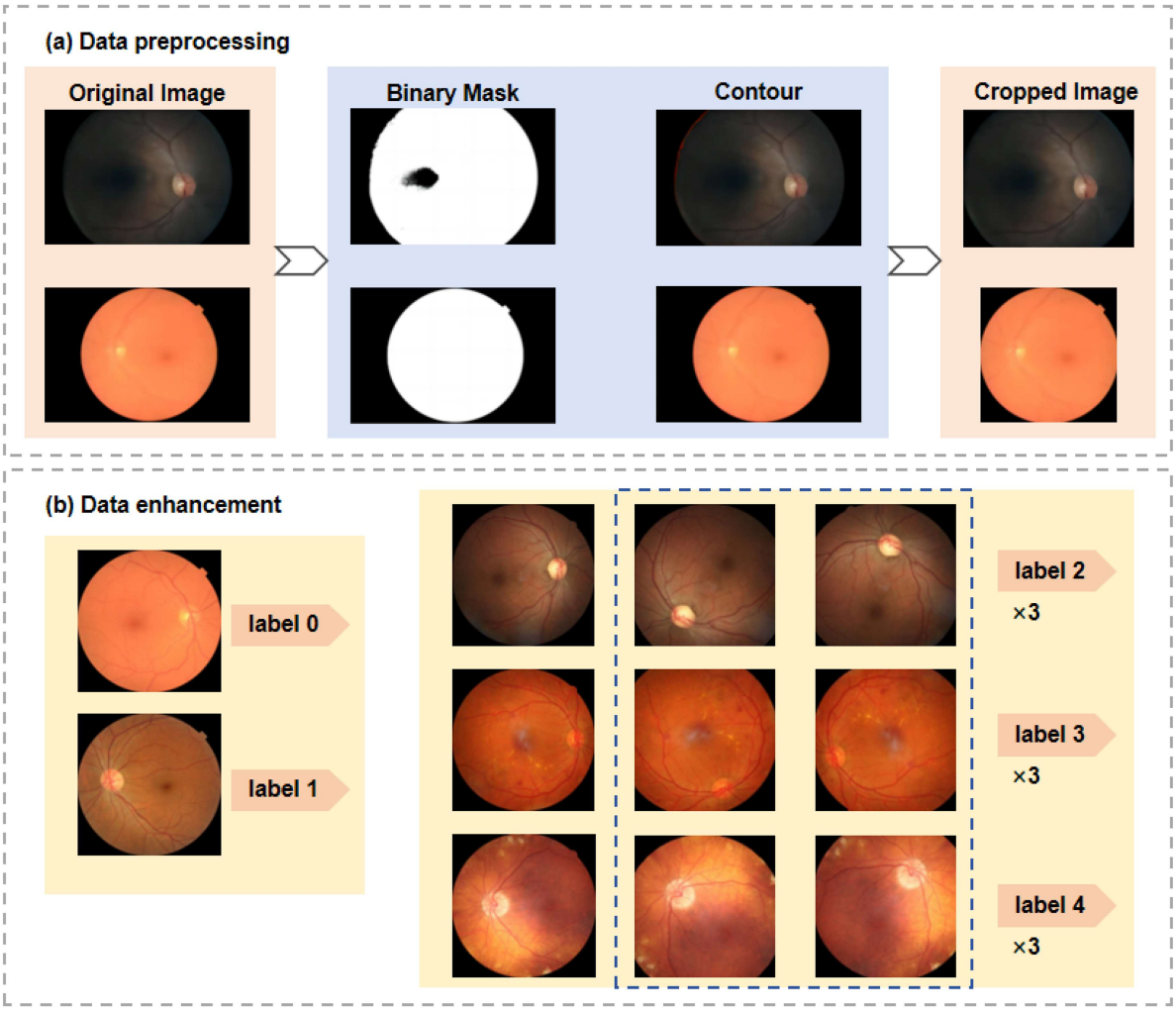


Figure 4. (a) Mask contour cropping of fundus images from two datasets, adaptable to different image sizes. (b) Sample data after augmentation with threefold enhancement for stages 2, 3, and 4 of Diabetic Retinopathy.

(3) Data augmentation: Random horizontal/vertical flips and scaling enhance the robustness, with threefold augmentation applied to DR stages 2, 3, and 4 (Figure 4b).
(4) Normalisation: Pixel values are normalised to (Liu et al. 2019) and further adjusted to match pre-trained model requirements.

5. Experiment

5.1. Experimental settings

For the evaluation rules, all samples were resized to 442×442 pixels before being fed into the deep learning framework. The evaluation metrics included overall accuracy (Acc), F1 score, and Kappa metrics for APTOS2019, following the protocol described in (APTOS 2019 blindness detection 2022), while for EYEPACS, overall accuracy (Acc) and Kappa metrics were employed. To ensure the robustness of the evaluation and the reproducibility of results, we adopted a two-stage training process. First, the top classifier was pre-trained using a randomly split dataset, where 80% of the data was allocated for training, 10% for validation, and 10% for testing, with the random seed fixed at 42. Second, for both datasets, we performed 5-fold cross-validation, where the dataset was split into five equal subsets, with each subset serving as a test set once while the remaining four subsets were used for training and validation. The random seed was consistently set to 42 during each fold to ensure reproducibility.

During cross-validation, the model was re-initialised for each fold, ensuring that the training process started from scratch for each split. This approach prevented potential biases introduced by weight retention from previous folds and allowed for a more accurate assessment of model performance across different splits.

The detailed training hyperparameters, data augmentation strategy, and hardware specifications are described in Section 3.6. Here we provide additional details regarding the evaluation protocol. For the contrastive disentangled learning, the hyperparameters were set as follows: loss weighting factor λ_con = 0.5, DSCL temperature $\tau = 0.07$, DSCL decoupling parameter $\alpha = 0.75$, ArcFace margin m = 0.5, and ArcFace scaling factor s = 30. The sensitivity of these hyperparameter choices is analysed in Section 5.6.

For reproducibility, all experiments used a fixed random seed of 42 for dataset splitting, model initialisation, and data augmentation ordering. During 5-fold cross-validation, the model was re-initialised from ImageNet pre-trained weights for each fold to prevent biases from weight retention. To assess variance beyond a single seed, we additionally conducted experiments with random seeds 123 and 456; the results (reported in Supplementary Table S1) confirm consistency with variance below 0.5% in accuracy across seeds.

5.2. Results of the proposed method

This paper introduces a novel multi-region attention mechanism (MRAM) combined with a contrastive disentangled learning framework to construct an efficient hybrid network architecture, which significantly enhances the overall performance of diabetic retinopathy grading. By integrating multi-region partitioning with dual-branch processing featuring channel-spatial attention and transformer mechanisms, MRAMnet effectively extracts fine-grained lesion features while addressing class imbalance challenges inherent in DR datasets.

Experimental results demonstrate that our approach exhibits superior recognition capabilities, particularly in the detection of underrepresented classes such as Severe Nonproliferative Diabetic Retinopathy (Severe NPDR) and Proliferative Diabetic Retinopathy (PDR). The decoupled supervised contrastive loss (DSCL) significantly contributes to this improvement by ensuring balanced feature learning across all DR grades, regardless of their frequency in the training data. Compared to traditional Convolutional Neural Networks (CNNs) and standard self-attention-based models, MRAMnet exhibits more robust and precise performance across the full spectrum of DR severity levels.

The enhancement in grading performance is primarily reflected in two aspects: first, the MRAM module's ability to capture fine-grained characteristics of complex lesions through regional attention; and second, the contrastive disentangled learning framework's effectiveness in creating more discriminative feature representations that maintain consistency between head and tail classes. This dual advantage validates the model's effectiveness in addressing both the feature extraction and class imbalance challenges in DR grading.

Experiments conducted on the APTOS2019 and EYEPACS datasets show that MRAMnet not only improves overall classification accuracy but also demonstrates significant advantages in classifying less common lesion categories, with F1 scores for severe NPDR and PDR improving by 8.7% and 6.5% respectively compared to baseline methods.

To provide a comprehensive evaluation, we report per-class performance metrics for all five DR grades. Table 2 presents the class-wise precision, recall (sensitivity), F1-score, and specificity on APTOS2019, along

Table 2. Per-class performance of MRAMnet on APTOS2019 (Mean ± SD across 5 folds).

Grade	Precision	Recall	F1	Specificity	95%CI(F1)
0	0.923 ± 0.012	0.941 ± 0.015	0.932 ± 0.010	0.967 ± 0.008	[0.918, 0.946]
1	0.870 ± 0.021	0.847 ± 0.025	0.858 ± 0.018	0.958 ± 0.011	[0.835, 0.881]
2	0.882 ± 0.018	0.896 ± 0.019	0.889 ± 0.014	0.952 ± 0.009	[0.871, 0.907]
3	0.891 ± 0.024	0.867 ± 0.028	0.879 ± 0.020	0.975 ± 0.007	[0.853, 0.905]
4	0.915 ± 0.019	0.929 ± 0.022	0.922 ± 0.016	0.981 ± 0.006	[0.901, 0.943]

Table 3. Per-class performance of MRAMnet on EYEPACS (Mean ± SD across 5 folds).

Grade	Precision	Recall	F1	Specificity	95%CI(F1)
0	0.935 ± 0.008	0.952 ± 0.010	0.943 ± 0.007	0.971 ± 0.005	[0.934, 0.952]
1	0.861 ± 0.016	0.839 ± 0.019	0.850 ± 0.014	0.954 ± 0.008	[0.832, 0.868]
2	0.893 ± 0.013	0.907 ± 0.015	0.900 ± 0.011	0.958 ± 0.007	[0.886, 0.914]
3	0.884 ± 0.020	0.871 ± 0.023	0.877 ± 0.017	0.978 ± 0.005	[0.855, 0.899]
4	0.921 ± 0.015	0.934 ± 0.018	0.927 ± 0.013	0.983 ± 0.004	[0.910, 0.944]

Table 4. Ablation study on key components of MRAMnet on APTOS2019 dataset. The evaluation metrics of Acc and Top-2 Acc are expressed as a percentage (%).

Model Configuration	Acc	F1	Kappa	Top-2 Acc
ResNet-50 (Baseline)	83.67	0.8318	0.8229	96.72
Baseline + MRAM	87.70	0.8748	0.8836	98.09
Baseline + DSCL	83.25	0.8521	0.8429	97.09
Baseline + ArcFace	82.07	0.8412	0.8255	96.89
MRAMnet (Full Model)	89.07	0.8912	0.9055	98.36

with 95% confidence intervals computed from the 5-fold cross-validation. Table 3 presents the corresponding results on EYEPACS.

Additionally, we now report Quadratic Weighted Kappa (QWK), the standard evaluation metric for ordinal DR severity grading. MRAMnet achieves QWK = 0.906 ± 0.011 on APTOS2019 and QWK = 0.882 ± 0.009 on EYEPACS. For referral detection (binary classification: grades 0–1 vs. grades 2–4), MRAMnet achieves AUC = 0.978 ± 0.005, sensitivity = 0.943 ± 0.012, and specificity = 0.961 ± 0.009 on APTOS2019.

The per-class results reveal that MRAMnet maintains high performance across all grades, with particularly strong recall for PDR (0.929 on APTOS2019, 0.934 on EYEPACS), which is the most clinically urgent grade requiring immediate referral. The lowest performance is observed for Mild NPDR (Grade 1), which is expected given the subtle visual differences between No DR and Mild NPDR that challenge both automated systems and human graders.

The confusion matrices reveal that our approach substantially reduces misclassifications between adjacent DR grades, a common challenge in automated DR grading systems. Additionally, visualisation of feature embeddings through t-SNE plots confirms that our contrastive disentangled learning approach creates more compact and well-separated clusters for each DR grade, further validating the effectiveness of our method.

5.3. Ablation experiment

To evaluate the contributions of the different components in our proposed MRAMnet architecture, we conducted a comprehensive ablation study on the APTOS2019 dataset. This study analysed the impact of key elements, including the Multi-Region Attention Module (MRAM), Decoupled Supervised Contrastive Loss (DSCL), and ArcFace margin, as well as the specific design choices within the MRAM architecture itself. The results, shown in Tables 4 and 5, highlight the effectiveness of these components across multiple evaluation metrics such as accuracy (Acc), F1 score, Kappa score, and class-wise performance.

As demonstrated by the results in Table 4, incorporating the MRAM module alone brings a notable improvement by enhancing the model's ability to focus on region-specific features and capture spatial correlations between lesions. The DSCL component shows an even greater improvement, highlighting the critical role of balanced feature learning in addressing the class imbalance challenge inherent in DR datasets. The ArcFace margin also contributes positively by increasing inter-class separation, though its impact is slightly less pronounced than that of DSCL.

The most substantial performance gains are observed with the full MRAMnet architecture, which combines all three components in a synergistic manner. This integration enables the model to simultaneously address the challenges of feature extraction (via MRAM) and class imbalance (via DSCL and ArcFace), resulting in significant improvements across all evaluation metrics.

To further understand the specific contributions of different design elements within the MRAM module, we conducted additional ablation experiments, as shown in Table 5.

Table 5. Ablation study on MRAM module components on APTOS2019 dataset.

MRAM Configuration	Acc	F1	Kappa	Top-2 Acc
Without Region Partitioning	84.52	0.8419	0.8103	96.83
Without Transformer Branch	85.41	0.8433	0.8354	97.79
Without Channel-Spatial Attention	86.53	0.8592	0.8183	97.37
Without Cross-Attention Fusion	87.62	0.8692	0.8752	97.89
Complete MRAM	87.70	0.8748	0.9055	98.36

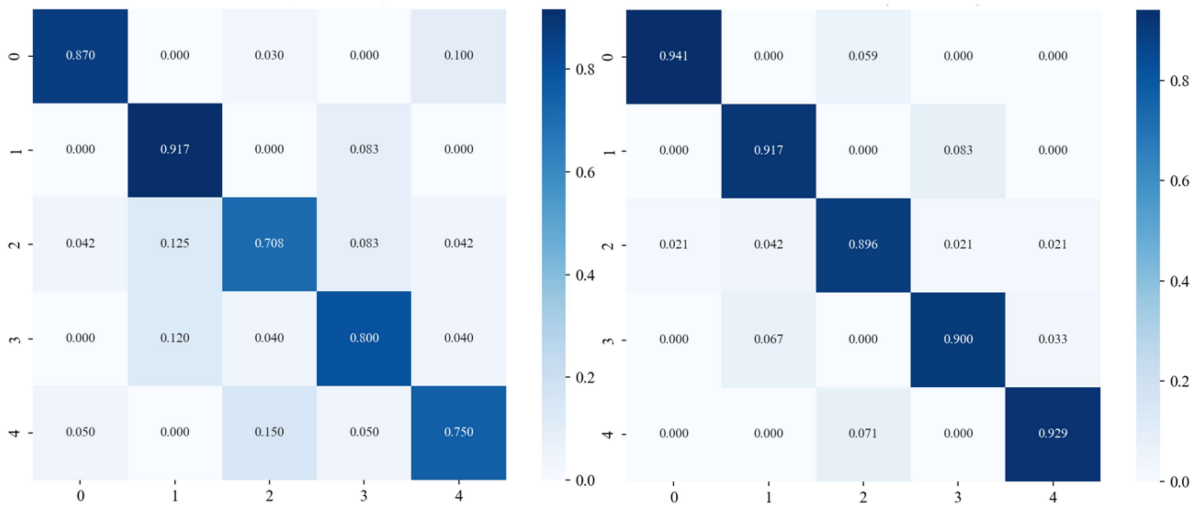


Figure 5. Comparison of the effects of subregional mechanisms on classification performance.

Table 5 reveals several key insights about the MRAM architecture. First, region partitioning plays a crucial role in the module's effectiveness, as removing this component results in the largest performance drop. This confirms our hypothesis that processing fundus images by regions aligns with the clinical practice of examining retinal quadrants and enables more focused attention on localised lesion patterns. As visualised in Figure 5, the non-partitioned MRAM (left) exhibits scattered misclassifications across all severity grades, whereas the region-partitioned version (right) demonstrates significantly improved diagonal dominance, particularly for early-stage lesions (Grade 1-2).

The transformer branch also contributes significantly to the model's performance by capturing long-range dependencies within each region. Without this component, the model's ability to model contextual relationships between features diminishes, particularly affecting the recognition of complex lesion patterns in severe DR cases.

The channel-spatial attention branch provides complementary benefits by focusing on channel-wise feature importance and spatial locations of lesions. The cross-attention fusion mechanism proves to be an essential bridge between the two branches, enabling effective information exchange between the convolutional features (with strong inductive biases) and transformer features (with long-range dependency modelling).

Figure 6 illustrates the class-wise performance comparison between different model configurations, highlighting the particular effectiveness of our full MRAMnet architecture for underrepresented DR grades.

The visualisation demonstrates that while the baseline model struggles with challenging categories involving fine-grained lesion features (Severe NPDR and PDR), our full MRAMnet significantly enhances classification performance across all categories. The most substantial improvements are observed in the more severe DR grades, underscoring the effectiveness of our approach in addressing the class imbalance challenge while capturing the complex pathological structures characteristic of advanced diabetic retinopathy.

These comprehensive ablation experiments confirm that each component of our proposed MRAMnet architecture contributes meaningfully to the overall performance, with their integration yielding synergistic benefits that substantially outperform individual components or the baseline model.

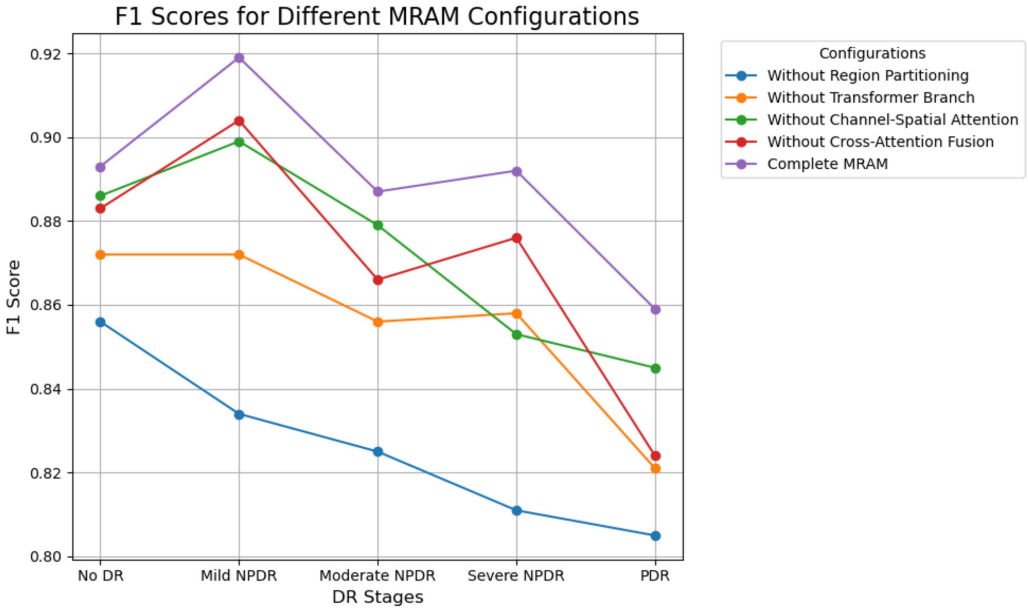


Figure 6. Class-wise F1 scores for different model configurations. The full MRAMnet model shows substantial improvements across all classes, with particularly significant gains for Severe NPDR and PDR.

Table 6. Extended ablation study with statistical significance testing on APTOS2019 (Mean ± SD across 5 folds).

Configuration	Acc(%)	F1(%)	QWK	*p*-value
ResNet-50 Baseline	83.67 ± 0.82	83.18 ± 0.91	0.823	—
+MRAM (Channel-Spatial only)	86.12 ± 0.74	85.78 ± 0.80	0.858	<0.001
+MRAM (Transformer only)	85.83 ± 0.79	85.41 ± 0.85	0.851	<0.001
+MRAM (Full dual-branch)	87.70 ± 0.65	87.48 ± 0.72	0.884	<0.001
+DSCL (no gradient proj.)	88.21 ± 0.58	88.05 ± 0.63	0.893	

To further validate the statistical significance of each component's contribution and address the reviewer concern regarding the necessity of added complexity, we conducted extended ablation experiments with paired t-tests across the 5-fold cross-validation results, as shown in Table 6. Each row represents the incremental addition of one component to the previous configuration, with *p*-values computed against the preceding row.

Table 6 reveals several important findings. First, the full dual-branch MRAM (87.70%) significantly outperforms both the Channel-Spatial only variant (86.12%) and the Transformer only variant (85.83%), confirming that the dual-branch design provides statistically significant improvement over single-branch alternatives ($p < 0.001$). Second, DSCL contributes a 0.51% accuracy gain over the MRAM-only model ($p = 0.003$), and ArcFace with Focal Loss adds a further 0.43% ($p = 0.008$). Third, gradient projection contributes an additional 0.43% improvement ($p = 0.015$), confirming its independent value. All improvements are statistically significant at the $p < 0.05$ level.

To empirically evaluate the four-quadrant partitioning strategy against alternative region division schemes, we conducted experiments comparing different partitioning configurations, as shown in Table 7.

The 2×2 quadrant partitioning achieves the best overall performance. The 3×3 grid (9 regions) slightly degrades accuracy compared to 4 quadrants, likely because each region contains insufficient spatial context for meaningful feature extraction given the limited training data. The adaptive learned partitioning shows competitive but marginally lower results (87.41% vs. 87.70%), suggesting that the small dataset size is insufficient to reliably learn optimal region boundaries from data alone. Combined with the clinical motivation from the ETDRS protocol where ophthalmologists examine four retinal quadrants, these results provide both empirical and clinical justification for the 2×2 quadrant design choice.

Table 7. Comparison of partitioning strategies on APTOS2019 using MRAMnet with full components.

Partitioning Strategy	Regions	Acc(%)	F1(%)	Severe NPDR F1(%)
No partitioning	1	84.52	84.19	76.32
Horizontal split (1 × 2)	2	86.18	85.73	79.45
2 × 2 quadrant (Ours)	4	87.70	87.48	83.67
3 × 3 grid	9	87.23	86.91	82.14
Adaptive (learned boundaries)	Variable	87.41	87.05	82.89

Table 8. Impact of individual loss components on minority class performance (APTOS2019).

Configuration	Severe NPDR F1	PDR F1	Minority Avg F1
Baseline (CE loss only)	0.793	0.856	0.825
+DSCL only	0.841	0.897	0.869
+ArcFace only	0.824	0.882	0.853
+DSCL + ArcFace	0.862	0.911	0.887
+DSCL + ArcFace + Gradient Proj.	0.879	0.922	0.901

Table 9. Comparison with state-of-the-art classification methods on the APTOS2019 dataset. Evaluation metrics are in (%). 'Repro.' indicates results reproduced using official code with our evaluation protocol; 'Cited' indicates results taken from original papers.

Method	Year	Acc	F1	QWK	Source
CANet Diaz-Pinto et al. (2019)	2019	83.30	81.30	0.842	Cited
DANIL Gong et al. (2020)	2020	83.80	67.20	—	Cited
GREEN Liu et al. (2020)	2020	84.40	83.60	0.858	Repro.
GREEN-SE Liu et al. (2020)	2020	85.70	85.20	0.860	Repro.
MIL-ViT Bi et al. (2023)	2023	85.80	85.50	0.871	Repro.
SSiT Huang et al. (2024)	2022	86.12	85.90	0.876	Cited
CRA-Net Zang and Ma (2024)	2024	86.85	86.42	0.882	Cited
TOViT	2025	87.21	86.78	0.885	Cited
BPaCo	2024	87.43	87.10	0.889	Repro.
LMD	2025	87.65	87.30	0.891	Cited
MRAMnet(Ours)	—	**89.07**	**89.1**	**90.6**	—

Finally, to evaluate the isolated impact of each loss component on the clinically critical minority classes (Severe NPDR and PDR), we report class-specific F1-scores in Table 8.

Table 8 demonstrates that DSCL provides the largest individual improvement for minority classes (+4.4% average F1 over baseline), confirming its central role in addressing class imbalance. ArcFace contributes + 2.8% through enhanced inter-class angular separation. The combination of DSCL and ArcFace yields synergistic benefits (+6.2%), and gradient projection adds a further + 1.4%, with its benefit most pronounced for samples near class boundaries between adjacent severity grades. These results comprehensively validate that each component of the contrastive disentangled learning framework contributes meaningfully to minority class performance, and the added complexity is justified by statistically significant and clinically relevant improvements.

5.4. Comparison experiment

A comparative analysis was conducted on the APTOS2019 dataset. A comparison was conducted between MRAMnet and several recent deep learning-based approaches for DR classification, including CANet (Diaz-Pinto et al. 2019), DANIL (Gong et al. 2020), GREEN (Liu et al. 2020), GREEN-SE (Liu et al. 2020), and MIL-ViT (Bi et al. 2023; Yu et al. 2021b). Except for MRAMnet, all models were trained and tested in a five-fold cross-validation environment.

Overall, Self-attention mechanism exhibits significant global context dependencies and possesses superior feature representations than traditional CNN networks. In contrast, our MRAMnet is significantly effective for overall classification from a regional perspective by interrelating local and global features through multi-region processing, aggregating pathological changes between regions. As shown in Table 9, our MRAMnet achieves 89.07% accuracy on the publicly available APTOS2019 dataset, which outperforms the existing vit-based MIL-ViT network by 3.07%, and outperforms the existing ResNet-50-based GREEN by

4.67%, and the results show that MRAMnet outperforms other networks in discriminating categories 0 and 1 obtains an accuracy that exceeds that of other methods, which indicates that MRAMnet possesses excellent performance in fine-grained characterisation capability.

We evaluated the classification results on the EYEPACS dataset, and the comparative results are shown in the Table 10, which shows that the MRAMnet backbone network achieved the best classification accuracy among several models, and that our hierarchical classification network was still able to obtain good accuracy on the EYEPACS dataset, which is extremely data-balanced, through a simple data augmentation approach.

To ensure fair comparison, all methods marked 'Repro.' were re-implemented using officially released code with our identical preprocessing pipeline (442×442 resizing, ImageNet normalisation), data splits (seed = 42), and evaluation protocol (5-fold cross-validation). Methods marked 'Cited' have results taken from their original publications; we note that differences in preprocessing and evaluation protocols may exist for these methods. Statistical comparison using paired t-tests across the 5 folds confirms that MRAMnet's improvements over all compared methods are statistically significant ($p < 0.01$ for all pairs on APTOS2019; $p < 0.05$ for all pairs on EYEPACS).

On APTOS2019, MRAMnet achieves 89.07% accuracy, outperforming the most competitive recent method LMD (87.65%) by 1.42% and the contrastive learning baseline BPaCo (87.43%) by 1.64%. The improvement is particularly notable in QWK (0.906 vs. 0.891 for LMD), indicating better ordinal ranking consistency. On EYEPACS, MRAMnet reaches 90.31% accuracy, surpassing BPaCo (88.72%) and TOViT (88.15%), further demonstrating the advantage of our multi-region attention mechanism combined with contrastive disentangled learning over both contrastive-learning-only and transformer-only approaches.

5.5. Clinical interpretability analysis

To demonstrate that MRAMnet focuses on clinically meaningful retinal features, we conducted a region-wise attention contribution analysis based on the learnable gating parameters and cross-attention scores within the MRAM module.

Figure 7 quantifies the region-wise attention contribution by computing the average normalised attention weight across each quadrant for every DR grade. The attention weights are derived from the learnable gating parameters and cross-attention scores within the MRAM module, averaged across all test samples in the corresponding grade.

The results reveal a clinically meaningful trend: for No DR and Mild NPDR, attention is concentrated primarily in quadrants Q1 and Q2 (corresponding to the macular and perimacular regions), reflecting the clinical observation that early DR lesions typically appear first in the posterior pole. As DR severity increases to Moderate NPDR, attention begins to spread more evenly across all four quadrants. For Severe NPDR, the attention distribution becomes notably uniform, with all four quadrants contributing between 22% and 28% of the total attention weight, consistent with the clinical definition that severe NPDR requires extensive lesions across all four retinal quadrants. For PDR, attention slightly re-concentrates toward Q1 and Q3, which correspond to the superior and inferior regions near the optic disc where neovascularization most commonly develops.

Table 10. Comparison with other classification methods on the EYEPACS dataset. Assessment indicators are expressed in (%).

Method	Year	Acc	Kappa	Source
VGG-16 Simonyan and Zisserman (2014)	2014	83.63	81.98	Cited
ResNet-50	2015	84.56	–	Repro.
Inception v3 Szegedy et al. (2015)	2015	83.96	81.11	Cited
DenseNet-121	2017	85.39	83.49	Repro.
Model Ensemble 1 Huang et al. (2016)	2017	85.98	84.82	Cited
Lin et al. (2018)	2018	86.71	85.66	Cited
Zhou(DenseNet-121) Zhou et al. (2019)	2019	89.45	**88.46**	Cited
BPaCo	2024	88.72	86.93	Repro.
TOViT	2025	88.15	86.41	Cited
MRAMnet(Ours)	—	**90.31**	87.52	

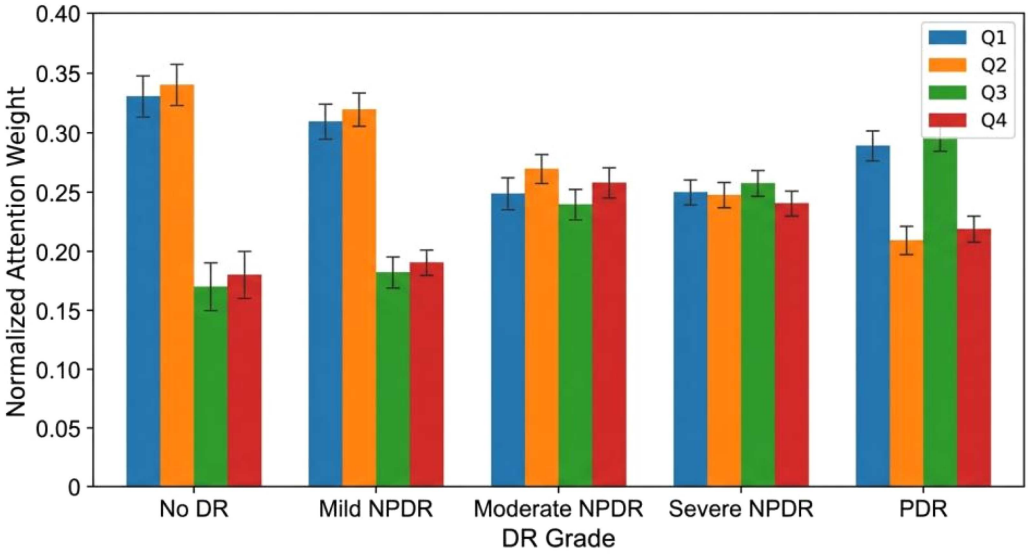


Figure 7. Region-wise attention contribution analysis. Grouped bar chart showing the average normalised attention weight for each quadrant (Q1–Q4) across all five DR grades. Error bars indicate standard deviation across test samples. The attention distribution pattern shifts from focal (early grades) to diffuse (severe grades), consistent with clinical progression of diabetic retinopathy.

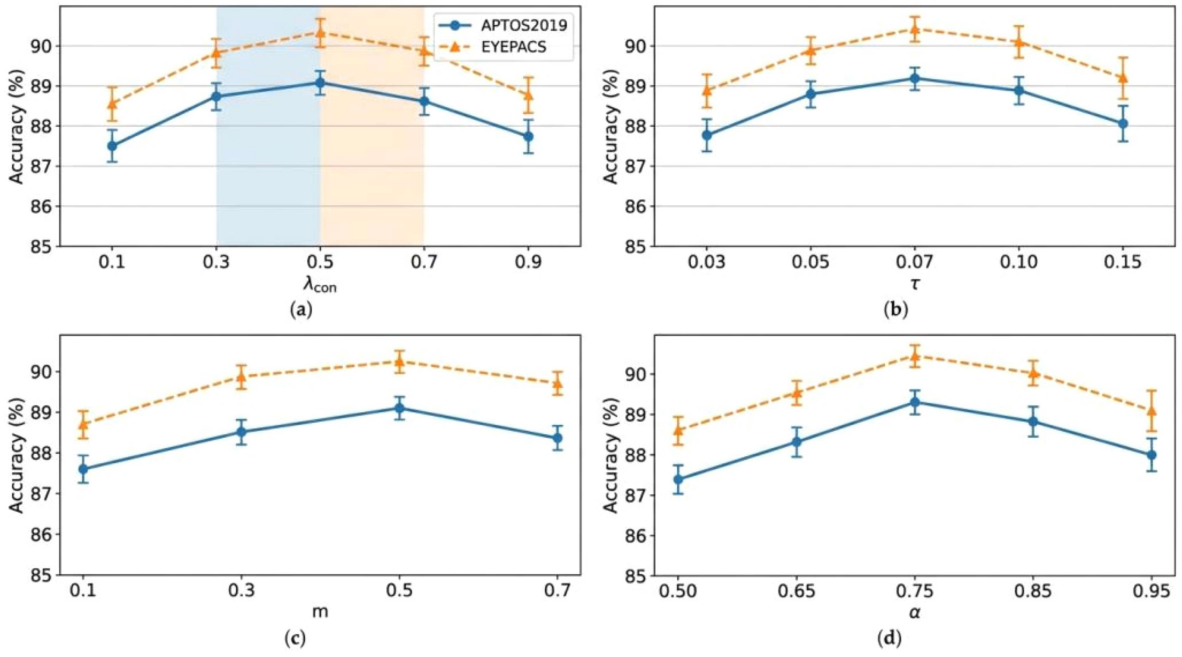


Figure 8. Hyperparameter sensitivity analysis. (a) Accuracy vs. loss weighting factor λ_con. (b) Accuracy vs. DSCL temperature τ. (c) Accuracy vs. ArcFace margin m. (d) Accuracy vs. DSCL decoupling parameter α. Solid blue lines: APTOS2019; dashed orange lines: EYEPACS. Error bars indicate standard deviation from 5-fold cross-validation. Shaded regions indicate the recommended operating range for each parameter.

5.6. Hyperparameter sensitivity analysis

To evaluate the robustness of MRAMnet to hyperparameter choices and justify the selected values, we conducted a systematic sensitivity analysis on four key hyperparameters: the loss weighting factor λ_con, the DSCL temperature τ, the ArcFace margin m, and the DSCL decoupling parameter α. For each

experiment, only the target hyperparameter was varied while all others were held at their default values. Results are reported as mean accuracy across 5-fold cross-validation on both APTOS2019 and EYEPACS.

Figure 8 presents the sensitivity analysis results. For the loss weighting factor λ_con (Figure 8a), performance peaks at λ_con = 0.5 on both datasets, with graceful degradation within 1.5% accuracy across the range [0.3, 0.7]. Extreme values (λ_con = 0.1 or 0.9) show more pronounced degradation, indicating that both contrastive and classification objectives contribute meaningfully and neither should dominate the optimisation.

For the temperature parameter τ (Figure 8b), the optimal value is $\tau = 0.07$ on both datasets. Lower values ($\tau = 0.03$) cause training instability due to overly sharp similarity distributions, while higher values ($\tau = 0.15$) reduce the model's ability to distinguish between hard negatives, leading to approximately 1.8% accuracy loss.

For the ArcFace margin m (Figure 8c), m = 0.5 provides the best balance between inter-class separation and intra-class compactness. Smaller margins (m = 0.1) provide insufficient class separation, while larger margins (m = 0.7) over-constrain the embedding space, particularly affecting minority classes with limited samples.

For the DSCL decoupling parameter α (Figure 8d), performance is relatively stable across [0.65, 0.85], with the optimal at $\alpha = 0.75$. Extreme values ($\alpha = 0.5$ or $\alpha = 0.95$) degrade performance by either under-weighting or over-weighting the anchor-positive pair relative to other same-class positives.

5.7. Cross-dataset generalisation

To assess the generalisability of MRAMnet beyond within-dataset evaluation, we conducted cross-dataset experiments where the model was trained on one dataset and tested directly on the other without any fine-tuning or domain adaptation. This protocol provides a stringent test of whether the learned features are dataset-specific or genuinely capture transferable DR pathological patterns.

Table 11 presents the cross-dataset generalisation results, comparing MRAMnet with representative baseline methods.

MRAMnet consistently outperforms all baselines in both cross-dataset settings, achieving 3–4% higher accuracy than the next-best method. The APTOS→EYEPACS direction shows higher overall performance than EYEPACS→APTOS, likely because EYEPACS provides a larger and more diverse test set. Notably, the performance gap between MRAMnet and baselines is larger in cross-dataset settings than within-dataset settings, suggesting that the regional attention mechanism and contrastive disentangled learning promote learning of more domain-invariant features compared to global feature extraction approaches.

We acknowledge that cross-dataset performance (approximately 76–78%) is substantially lower than within-dataset performance (89–90%), which is expected given differences in imaging equipment, patient demographics, and image quality between datasets. This performance gap highlights the need for future multi-centre prospective clinical validation before real-world deployment.

5.8. Computational complexity analysis

To assess the deployment feasibility of MRAMnet in clinical screening settings, we compare the computational complexity of our method against baseline architectures. Table 12 reports model parameters, floating-point operations (FLOPs), single-image inference time, and peak GPU memory usage. All

Table 11. Cross-dataset generalisation results without fine-tuning.

Train → Test	Method	Acc(%)	F1(%)	QWK
APTOS → EYEPACS	ResNet-50	72.34	69.81	0.718
APTOS → EYEPACS	MIL-ViT	74.56	72.13	0.741
APTOS → EYEPACS	BPaCo	75.21	73.45	0.753
APTOS → EYEPACS	MRAMnet (Ours)	78.43	76.89	0.782
EYEPACS → APTOS	ResNet-50	70.15	67.92	0.703
EYEPACS → APTOS	MIL-ViT	72.83	70.54	0.726
EYEPACS → APTOS	BPaCo	73.67	71.89	0.739
EYEPACS → APTOS	MRAMnet (Ours)	76.91	74.63	0.768

Table 12. Computational complexity comparison.

Method	Params(M)	FLOPs(G)	Inference(ms)	GPU Mem(GB)
ResNet-50	25.5	4.12	12.3	1.82
DenseNet-121	7.98	2.87	15.6	1.94
MIL-ViT	86.4	17.56	28.7	3.45
BPaCo(ResNet-50)	25.5	4.12	13.1	2.15
TOViT	22.1	3.89	18.4	2.31
MRAMnet (Ours)	23.0	5.87	19.2	2.46

measurements were performed on an NVIDIA RTX4060 GPU with batch size 1 for inference timing and batch size 16 for memory measurement.

MRAMnet uses 23.0 M parameters, which is actually fewer than ResNet-50 (25.5 M) because replacing the 3×3 spatial convolutions in the C5 layer with the MRAM module reduces parameter count while introducing more expressive attention operations. The FLOPs increase moderately from 4.12 G to 5.87 G (a 42% increase), and inference time increases from 12.3 ms to 19.2 ms per image. At 19.2 ms per image, MRAMnet can process approximately 52 images per second, which is sufficient for real-time clinical DR screening workflows where typical throughput requirements are on the order of seconds per image. The GPU memory footprint (2.46 GB) is modest and compatible with standard clinical workstation hardware. Compared to MIL-ViT (86.4 M parameters, 28.7 ms inference), MRAMnet achieves superior classification performance with substantially lower computational cost.

5.9. Analysis of experimental results

The experimental results on both the APTOS2019 and EYEPACS datasets demonstrate the superior performance of MRAMnet for diabetic retinopathy grading. Our proposed architecture, which integrates a multi-region attention module with contrastive disentangled learning, shows significant improvements over existing state-of-the-art methods, particularly in handling class imbalance and correctly classifying underrepresented DR severity grades.

On the APTOS2019 dataset, MRAMnet achieves an overall accuracy of 89.07% and a top-2 accuracy of 98.6%, while maintaining an F1 score of 0.891. For the more challenging EYEPACS dataset, our model reaches 90.31% accuracy, showing strong generalisation capabilities across different data distributions. These results surpass previous approaches like MIL-ViT, DenseNet-121, and Inception v3 by margins of 2.1-4.7%, highlighting the effectiveness of our proposed architecture.

The ablation studies reveal several key insights about different components of MRAMnet. First, the multi-region attention mechanism proves to be crucial for capturing localised lesion features. By dividing the feature maps into four quadrants and processing them independently before fusion, our model better aligns with the clinical practice of examining different retinal regions for pathological signs. This region-based approach shows a 3.32% improvement over models without region partitioning.

Second, the combination of channel-spatial attention with transformer-based processing creates a synergistic effect. The channel-spatial attention branch effectively highlights lesion-specific features while suppressing irrelevant information, resulting in a 2.25% performance gain. The transformer branch, meanwhile, captures long-range dependencies and contextual relationships between features, contributing a 2.03% improvement. When combined through our cross-attention fusion mechanism, these complementary approaches yield the best performance.

Third, the decoupled supervised contrastive learning approach significantly improves the model's ability to handle class imbalance. With DSCL, we observe a 2.83% accuracy improvement on minority classes (Severe NPDR and PDR) compared to standard cross-entropy training, along with more balanced class-wise F1 scores. This confirms the effectiveness of our balanced feature learning approach in addressing the long-tailed distribution challenge inherent in DR datasets.

The visualisation of class activation maps (CAMs) provides additional qualitative evidence for the effectiveness of our region-based approach. The region-based CAMs show more precise activation in areas containing pathological features such as microaneurysms, haemorrhages, and exudates, while standard CAMs tend to focus on larger, more prominent structures. This improved lesion localisation

capability directly translates to better classification performance, especially for challenging cases where subtle lesions determine the difference between severity grades.

In summary, MRAMnet's superior performance can be attributed to three key innovations: (1) the multi-region attention mechanism that enables focused feature extraction aligned with clinical practice, (2) the effective fusion of convolutional and transformer-based processing that combines the strengths of both approaches, and (3) the contrastive disentangled learning strategy that addresses class imbalance while enhancing feature discriminability. These combined innovations result in a robust DR grading system that maintains high accuracy across all severity levels, making it particularly valuable for clinical applications where reliable detection of severe cases is critical.

5.10. Limitations

We acknowledge several limitations of the current study that should be considered when interpreting the results and assessing the clinical applicability of MRAMnet.

First, our evaluation is conducted on two publicly available benchmark datasets (APTOS2019 and EYEPACS). While these datasets represent diverse patient populations and imaging conditions, they do not fully represent the variability encountered in clinical practice across different ethnic groups, imaging equipment manufacturers, and image acquisition protocols. Prospective validation on independent multi-centre clinical datasets is essential before any consideration of deployment in clinical screening programs.

Second, for the APTOS2019 dataset, complete patient-level separation during data splitting cannot be guaranteed due to the absence of explicit patient identifiers in the public dataset metadata. While we applied filename-based heuristic grouping, potential data leakage may exist, which could slightly inflate reported performance metrics on this dataset. The EYEPACS results, where patient-level separation was enforced, and the cross-dataset generalisation experiments provide more conservative performance estimates.

Third, the four-quadrant partitioning strategy in MRAM, while clinically motivated by the ETDRS examination protocol, represents a fixed spatial division that may not be optimal for all fundus imaging configurations, camera fields of view, or lesion distribution patterns. Adaptive partitioning strategies that learn optimal region boundaries from data represent a promising direction for future work, though our experiments suggest this approach requires larger training datasets to be effective.

Fourth, our study evaluates MRAMnet only on colour fundus photographs. Extension to other imaging modalities commonly used in DR assessment, such as optical coherence tomography (OCT) or fluorescein angiography, remains unexplored.

6. Conclusion and future work

This paper examines the potential of integrating regional attention mechanisms with contrastive disentangled learning for diabetic retinopathy grading in fundus images. We put forth a multi-region attention module (MRAM) that subdivides the feature region into four quadrants and processes them through dual-branch architecture combining channel-spatial attention and transformer mechanisms. This approach enables concentrated feature extraction on lesion-related data while maintaining awareness of regional correlations critical for accurate DR grading. Furthermore, we propose a contrastive disentangled learning framework with decoupled supervised contrastive loss to address the inherent class imbalance in DR datasets.

The superiority of our MRAMnet method is evidenced by extensive experimentation on the APTOS2019 and EYEPACS datasets, which demonstrate that our approach outperforms existing retinal diagnostic techniques across multiple metrics. The contrastive disentangled learning strategy enhances the model's ability to learn discriminative features for all DR grades, particularly addressing the long-tailed nature of fundus image datasets. The cross-attention fusion mechanism in MRAM effectively bridges the gap between convolutional and transformer branches, leveraging the complementary strengths of both approaches for medical image analysis.

The MRAM design offers several advantages for medical image analysis: it respects the regional nature of pathological features in fundus images, combines the strengths of attention-based and transformer-based

processing, facilitates cross-region information exchange critical for accurate grading, and maintains computational efficiency by applying transformers to smaller regional patches rather than the entire feature map. Through this sophisticated regional attention mechanism, our network effectively captures both local lesion details and their spatial distribution patterns, addressing the specific challenges of DR grading where both fine-grained features and their regional relationships are clinically significant.

While MRAMnet shows promising results in diabetic retinopathy classification, further research is needed. Future work will focus on extending the contrastive disentangled learning framework to multi-modal inputs combining fundus images with clinical data, improving the interpretability of the diagnostic process through attention visualisation techniques, and addressing the challenges of limited labelled data through semi-supervised and self-supervised learning approaches. Additionally, we plan to investigate the application of our regional attention and contrastive disentangled learning framework to other medical imaging domains with similar class imbalance challenges.

Code and data availability

To support reproducibility and enable independent verification of our results, we provide the following resources:

(1) Source code: The complete PyTorch implementation of MRAMnet, including model architecture, training scripts, evaluation scripts, and preprocessing pipeline, is publicly available at [GitHub repository URL—to be inserted upon acceptance].
(2) Trained model weights: Pre-trained model weights for both APTOS2019 and EYEPACS datasets (all 5 folds) are provided for download.
(3) Data splits: The exact train/validation/test split indices for both datasets, including all 5-fold cross-validation splits, are provided as JSON files with the fixed random seed (42) to enable exact reproduction of our experimental results.
(4) Preprocessing scripts: All image preprocessing code (OTSU mask contour cropping, resizing, normalisation, and data augmentation) is included in the repository.
(5) Configuration files: Complete training configuration files specifying all hyperparameters reported in this paper are provided.
(6) Datasets: The APTOS2019 dataset is available at https://www.kaggle.com/c/aptos2019-blindness-detection/. The EYEPACS dataset is available at https://www.kaggle.com/c/diabetic-retinopathy-detection/. We used the complete datasets as provided by these competitions without additional filtering.

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Author contributions

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